

Three Consecutive Square-Pyramidal Numbers

Exact Frobenius Formulas, Gap Laws, and Algebraic Inversion

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Abstract

Let $s_n = n(n+1)(2n+1)/6$ and let F_n be the largest integer not in $\langle s_n, s_{n+1}, s_{n+2} \rangle$. This is OEIS A069762. We give an explicit solution: F_n is a degree-five quasipolynomial of least eventual period six, with exactly six exceptional indices, the last being 23. In particular, $F_n = n^5/9 + 8n^4/9 + O(n^3)$, and all subsequent terms are given exactly.

The proof does considerably more than evaluate a maximum. Two elementary gcd reductions expose a linear-size Apéry set controlled by the relation $kC - \ell A = 5B$, with $k, \ell \leq 6$. This yields a direct membership and representation algorithm, exact genus and symmetry formulas, and exact quasipolynomial formulas for every fixed power sum of the gaps. A general transfer principle converts limits of normalized Apéry sets into limits of gap measures. Applied here, a uniformly chosen gap divided by F_n converges to the density $2(1-x)$ on $[0, 1]$, with a uniform $O(n^{-1})$ error in the cumulative distribution. The ordinary generating function has minimal denominator $(1-x)^6(1+x)^4(1+x+x^2)^4$, hence a minimal eventual recurrence of order 18.

Finally, each residue-class polynomial admits a convergent all-orders Puiseux inverse. We give its first coefficients, a coefficient-extraction formula, a uniform sufficient convergence region, and an exact rule for recovering the discrete inverse. All main claims are proved in the text; exact computational checks and symbolic moment-generation programs accompany the report. The explicit formulas are proposed as additions to the inspected OEIS entry; literature priority is not certified.

Scope of the claim. This is an independently derived research report, not a claim that a famous named conjecture has been settled. General eventual quasipolynomiality is established literature. The contribution is the explicit formula, complete normal form, quantitative gap law, and effective consequences for this particular canonical family. No Lean or Rocq formalization of these results is claimed.

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1 The target, the literature, and the proof boundary

The inspected version of OEIS A069762 defines the Frobenius numbers generated by three consecutive square-pyramidal numbers, supplies values for $n = 2, \dots, 35$, and gives the example $F(5, 14, 30) = 51$. It contains no explicit general formula or asymptotic formula [1]. Its displayed definition is not labelled a conjecture. Thus the precise problem addressed here is to supply a missing explicit solution and its proof, rather than to reclassify an OEIS omission as a well-established open conjecture.

This is a natural parametric Frobenius problem, not an artificial sequence chosen solely because a few terms interpolate. Frobenius numbers for three consecutive squares and cubes were explicitly analyzed by Lepilov, O’Rourke, and Swanson [2]. Related work of Robles-Pérez and Rosales studies consecutive triangular and tetrahedral numbers [3]. Those families provide relevant context, but square-pyramidal numbers are a different polynomial family. Roune and Woods proved eventual quasipolynomiality for parametric Frobenius problems with at most three generators [4]; Shen established the broader polynomial-parametric result [5]. We do not claim eventual quasipolynomiality itself as new.

The source check for this report used the OEIS entry, the cited primary literature, searches for A069762 and for Frobenius numbers of square-pyramidal numbers, and a scoped inspection of the ProveIt repository. No publication containing the formula below was located. This is a bounded literature check, not a proof of priority. A prior equivalent formula, perhaps under different terminology, could change the novelty assessment without changing the mathematical statements proved here.

The connection to ProveIt is methodological rather than a dependency on an unverified repository theorem. Its README separates exploratory computations from the trusted boundary of formal mathematics [6]. We follow that separation: computation suggested the formula, but the proof is the explicit normal form developed below. The accompanying programs provide independent finite checks and algebraic audits. They are neither a proof of literature novelty nor a kernel-checked formalization. The repository snapshot used for this inspection is recorded in the bibliography; the repository was not modified.

[write] **Added 2 October 2026, batch 75.** This report is a single manuscript, printed in full: manuscript 02 of batch 75 of ProveIt’s incoming-reports intake, delivered as the archive `OEIS_Pyramidal_Frobenius.zip` in commit `4b874cea0` and placed in the research-report collection, among the enumerative-combinatorics reports, in commit `6ea60e367`. The snapshot named in the bibliography [6], there called a “root-tree snapshot”, is the ProveIt `commit 29aca108e` (1 October 2026), an ancestor of the placement commit; it is this manuscript’s pin. No ProveIt Lean or Rocq file treats Frobenius numbers or numerical semigroups, and nothing in this report is formalized; placement in the collection confers no formal status on any statement here. The writing step changed only the labels (every label now carries the prefix `pyf:`), added these [write] notes and two references [7, 8]; no statement, proof or number of the manuscript was altered. Eventual quasipolynomiality is the theorem of Roune–Woods and Shen [4, 5], as the paragraph above says; the contribution claimed is the explicit formula and its consequences. The nearest collection report, `numerical-semigroup-leaf-types` [8], studies the type of leaves of the numerical-semigroup tree (a genus-34 counterexample to a conjecture of Chappelon, Ramírez Alfonsín and Stamate); it treats no parametric family, and the two reports share no theorem. The collection report `constrained-crossover-closure` uses only the two-generator Frobenius number, as a lemma.

What is established here. The main output is an exact answer for *every* $n \geq 2$, not just a fitted asymptotic equivalent. The proof also determines all missing integers through a constant-description family of residue representatives. The distribution and moment results explain the global shape of the gaps, while the inverse results address the associated threshold problem. These additional results make the normal form reusable beyond a single OEIS formula.

2 An exact formula for A069762

For $n \geq 2$, put

$$s_n = \sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}, \quad S_n = \langle s_n, s_{n+1}, s_{n+2} \rangle, \quad F_n = \max(\mathbb{N}_0 \setminus S_n). \quad (2.1)$$

Here $\langle a, b, c \rangle = \{ua + vb + wc : u, v, w \in \mathbb{N}_0\}$. Define the periodic integer

$$h_n = 3\mathbf{1}_{\{n \text{ odd}\}} + 2\mathbf{1}_{\{n \equiv 2 \pmod{3}\}}. \quad (2.2)$$

Its values for residues 0, 1, 2, 3, 4, 5 modulo six are 0, 3, 2, 3, 0, 5. For $h \in \{0, 2, 3, 5\}$ define

$$P_h(X) = \frac{1}{36}(4X^5 + 32X^4 + (101 - 10h)X^3 + (175 - 45h)X^2 + (102 - 65h)X - 36 - 30h). \quad (2.3)$$

Equivalently,

$$P_h(X) = P_0(X) - \frac{5h}{36}(X+1)(X+2)(2X+3). \quad (2.4)$$

Theorem 2.1 (Exact Frobenius formula). *For every integer $n \geq 2$,*

$$\boxed{F_n = P_{h_n}(n) + \varepsilon_n}, \quad (2.5)$$

where $\varepsilon_n = 0$ except at the six indices in Table 2. Consequently $F_n = P_{h_n}(n)$ for all $n \geq 24$, and 24 is the least uniform cutoff for this quasipolynomial. Its least eventual period is six.

Table 1: The four polynomial branches. The common numerator begins $4n^5 + 32n^4$ and is divided by 36.

$n \bmod 6$	h_n	coefficient of n^3	of n^2	of n	constant
0, 4	0	101	175	102	-36
1, 3	3	71	40	-93	-126
2	2	81	85	-28	-96
5	5	51	-50	-223	-186

Table 2: All exceptions: the correction is positive in each case.

n	$P_{h_n}(n)$	ε_n	F_n
2	41	10	51
3	151	40	191
5	1009	315	1324
11	32553	1612	34165
17	238451	3135	241586
23	980251	2400	982651

The exact formula immediately yields

$$F_n = \frac{n^5}{9} + \frac{8n^4}{9} + \frac{101 - 10h_n}{36}n^3 + \frac{175 - 45h_n}{36}n^2 + \frac{102 - 65h_n}{36}n - 1 - \frac{5h_n}{6} \quad (2.6)$$

for $n \geq 24$. This is a complete finite asymptotic expansion with periodic coefficients: there is no remainder after the constant term. On each residue class it is simply a polynomial identity. In particular, $F_n \sim n^5/9$. A single constant coefficient at order n^3 would be incorrect; the first oscillatory term occurs at precisely that order.

The exact formula and the exception list are proved in Sections 3–5. Minimality of the period is proved in Section 9.

3 Apéry sets and two elementary reductions

3.1 The residue description of all gaps

A numerical semigroup is an additive submonoid of $\mathbb{N}_0$ with finite complement. Positive integers with gcd one generate such a semigroup: modulo any one generator, their residues generate the entire finite cyclic group, and a submonoid of a finite group is a subgroup. Thus each residue has a nonnegative representative; adding multiples of the chosen generator proves cofiniteness.

For a numerical semigroup S and $m \in S \setminus \{0\}$, let w_r be the least element of S congruent to r modulo m , for $0 \leq r < m$. The Apéry set is

$$\text{Ap}(S; m) = \{w_0, \dots, w_{m-1}\}.$$

In residue r , the gaps are exactly

$$r, r + m, \dots, w_r - m, \quad (3.1)$$

with the list empty when $w_r = r$. In particular $w_0 = 0$, so zero is never counted as a gap. It follows directly that

$$F(S) = \max_r w_r - m, \quad g(S) = \#(\mathbb{N}_0 \setminus S) = \frac{1}{m} \sum_r w_r - \frac{m-1}{2}, \quad (3.2)$$

and, as a formal power series,

$$H_S(z) := \sum_{s \in S} z^s = \frac{\sum_{w \in \text{Ap}(S; m)} z^w}{1 - z^m}. \quad (3.3)$$

These facts are standard in numerical-semigroup theory; the short derivation above fixes all indexing conventions used in this report. The gcd reduction below is the mechanism behind the classical Johnson Frobenius reduction recalled in [3], but we prove the full normal form needed here.

3.2 A rectangular lifting lemma

Lemma 3.1 (Two-stage normal form). *Let a, b, c be positive integers with gcd one. Set*

$$d = \gcd(a, b), \quad e = \gcd(b, c), \quad A = a/d, \quad B = b/(de), \quad C = c/e.$$

Then d, e are coprime, the displayed quantities are positive integers, and with $T = \langle A, B, C \rangle$ one has the disjoint decomposition

$$\langle a, b, c \rangle = \bigsqcup_{\substack{0 \leq i < e \\ 0 \leq j < d}} (ia + jc + deT). \quad (3.4)$$

The offsets give all residues modulo de . Moreover,

$$\text{Ap}(\langle a, b, c \rangle; b) = \{de w + ia + jc : w \in \text{Ap}(T; B), 0 \leq i < e, 0 \leq j < d\}. \quad (3.5)$$

Consequently,

$$F(\langle a, b, c \rangle) = de F(T) + (e-1)a + (d-1)c, \quad (3.6)$$

$$g(\langle a, b, c \rangle) = de g(T) + \frac{(e-1)a + (d-1)c - de + 1}{2}. \quad (3.7)$$

Proof. Any common divisor of d, e divides a, b, c , hence $\gcd(d, e) = 1$ and $de \mid b$. Also $\gcd(d, c) = \gcd(e, A) = 1$. In a representation $ua + vb + wc$, reduce the coefficient w modulo d . This gives uniquely a residue $0 \leq j < d$ and a remainder in $d\langle A, eB, c \rangle$. In that remainder reduce the coefficient of A modulo e . This gives $0 \leq i < e$ and a remainder in $e\langle A, B, C \rangle$. Conversely, every expression on the right of (3.4) is a nonnegative combination of a, b, c , since $deA = ea$, $deB = b$, and $deC = dc$.

If two offsets are equal modulo de , reduction modulo d first forces their j values to agree, because c is invertible modulo d . After division by d , reduction modulo e forces their i values to agree, because A is invertible modulo e . This proves disjointness and shows that all de residues occur. For each fixed

offset, minimizing further modulo $b = deB$ is exactly the Apéry minimization in T modulo B . This proves (3.5).

Taking the maximum in (3.5) proves (3.6). Its sum is

$$(de)^2 \sum_{w \in \text{Ap}(T; B)} w + Bde \left(\frac{e-1}{2}a + \frac{d-1}{2}c \right).$$

Apply (3.2), with modulus deB , to obtain (3.7). $\square$

3.3 The gcds of consecutive square-pyramidal numbers

Throughout the remainder of the paper, unless otherwise specified,

$$\begin{aligned} a &= s_n, & b &= s_{n+1}, & c &= s_{n+2}, \\ k &= \gcd(n+1, 6), & \ell &= \gcd(n+2, 6), & d &= \frac{n+1}{k}, & e &= \frac{n+2}{\ell}, \\ A &= \frac{kn(2n+1)}{6}, & B &= \frac{k\ell(2n+3)}{6}, & C &= \frac{\ell(n+3)(2n+5)}{6}. \end{aligned} \quad (3.8)$$

The relation between these symbols and Lemma 3.1 is exact.

Lemma 3.2. *One has $\gcd(a, b) = d$, $\gcd(b, c) = e$, and $\gcd(a, b, c) = 1$. The values of k, ℓ, B are given by Table 3. In particular, $\gcd(k, \ell) = 1$, B is odd, and $B \equiv 1 \pmod{k}$.*

Proof. For $m = n + 1$,

$$\gcd(s_{m-1}, s_m) = \gcd\left(\frac{(m-1)m(2m-1)}{6}, m^2\right).$$

If a prime $p \geq 5$ divides m , neither $m-1$ nor $2m-1$ is divisible by p , so the p -adic valuation of this gcd is $v_p(m)$. If $2 \mid m$, the other two factors are odd and the valuation is $v_2(m) - 1$; if $3 \mid m$, the valuation is $v_3(m) - 1$. Primes not dividing m do not contribute. Thus the gcd is $m / \gcd(m, 6)$, proving the two adjacent-gcd formulas. A common divisor of a, b, c divides both $b - a = (n+1)^2$ and $c - b = (n+2)^2$, which are coprime. The table is obtained by evaluating two gcds modulo six. Its remaining assertions follow at once. $\square$

Table 3: The reduced parameters. The linear modulus B replaces the original cubic modulus b .

$n \bmod 6$	k	ℓ	B	gaps of $\langle k, \ell \rangle$
0	1	2	$(2n+3)/3$	none
1	2	3	$2n+3$	$\{1\}$
2	3	2	$2n+3$	$\{1\}$
3	2	1	$(2n+3)/3$	none
4	1	6	$2n+3$	none
5	6	1	$2n+3$	none

The decisive identities are

$$\boxed{kC - \ell A = 5B}, \quad 2A \equiv k \pmod{B}, \quad 2C \equiv \ell \pmod{B}. \quad (3.9)$$

For completeness, exact forms of the congruences are

$$2A - k = B \frac{2(n-1)}{\ell}, \quad 2C - \ell = B \frac{2(n+4)}{k}. \quad (3.10)$$

Both quotients are integers in each row of Table 3. The first identity follows by subtracting the two quadratic expressions in (3.8). No approximation is involved in these reductions.

4 A bounded-coefficient Apéry theorem

4.1 A reusable reduction principle

Lemma 4.1 (Least auxiliary residue principle). *Let A, B, C, k, ℓ, H be positive integers with $\gcd(k, \ell) = 1$. Suppose*

$$kC - \ell A = HB, \quad tA \equiv k \pmod{B}, \quad tC \equiv \ell \pmod{B}, \quad \gcd(t, B) = 1.$$

Assume $A > H(k-1)$. For each residue modulo B , take the least nonnegative member L of $\langle k, \ell \rangle$ in that residue. Let

$$0 \leq v(L) < k, \quad \ell v(L) \equiv L \pmod{k}, \quad u(L) = \frac{L - \ell v(L)}{k}.$$

Then

$$\text{Ap}(\langle A, B, C \rangle; B) = \{Au(L) + Cv(L)\} = \left\{ \frac{AL + HBv(L)}{k} \right\}, \quad (4.1)$$

where L runs through the selected B auxiliary representatives.

Proof. Every candidate for a least residue representative can be written using A, C only, because a positive coefficient of B can be removed. Replacing kC by ℓA lowers the value by HB without changing its residue modulo B . Thus a minimizing representation has its coefficient of C in $[0, k-1]$.

For a normalized representation $uA + vC$, set $L = ku + \ell v$. The congruence class of L modulo B is $t(uA + vC)$, so auxiliary residues and value residues correspond bijectively. For any representable L , the normalized coefficient $v(L)$ is unique, and its coefficient $u(L)$ is nonnegative: normalize any nonnegative representation of L in $\langle k, \ell \rangle$ by reducing its coefficient of ℓ modulo k . The relation gives the second formula in (4.1).

Let L be the least representable auxiliary integer in a fixed residue. Every other representable auxiliary integer there is $L + qB$, with $q \geq 1$. The corresponding difference of normalized weights is

$$\frac{B}{k} (Aq + H(v(L + qB) - v(L))) \geq \frac{B}{k} (A - H(k-1)) > 0.$$

Hence the least auxiliary integer gives the least value. There is one such integer in every residue because $\langle k, \ell \rangle$ has gcd one. This proves both completeness and minimality. $\square$

The point of this lemma is not merely to reduce the number of generators. It replaces an unbounded coefficient search by a bounded coefficient $0 \leq v < k$ and a least-residue problem in a fixed, very small semigroup. The constant H need not equal five, and the parameter family need not be square-pyramidal.

4.2 The complete reduced Apéry set

For the parameters (3.8), define

$$\mathcal{L}_n = \begin{cases} (\{0, 1, \dots, B-1\} \setminus \{1\}) \cup \{B+1\}, & n \equiv 1, 2 \pmod{6}, \\ \{0, 1, \dots, B-1\}, & \text{otherwise.} \end{cases} \quad (4.2)$$

For $L \in \mathcal{L}_n$, put

$$\begin{aligned} v(L) &\in \{0, \dots, k-1\}, & \ell v(L) &\equiv L \pmod{k}, \\ u(L) &= \frac{L - \ell v(L)}{k}, & w(L) &= Au(L) + Cv(L) = \frac{AL + 5Bv(L)}{k}. \end{aligned} \quad (4.3)$$

When $k = 1$, the definition means simply $v(L) = 0$.

Theorem 4.2 (Explicit Apéry normal form). *For every $n \geq 2$,*

$$\text{Ap}(\langle A, B, C \rangle; B) = \{w(L) : L \in \mathcal{L}_n\}, \quad (4.4)$$

and

$$\boxed{\text{Ap}(S_n; b) = \{de w(L) + ia + jc : L \in \mathcal{L}_n, 0 \leq i < e, 0 \leq j < d\}}. \quad (4.5)$$

All the representatives in (4.5) have different residues modulo b . Thus this formula gives every gap by (3.1).

Proof. Table 3 shows that the only missing auxiliary integer is 1, precisely in the two rows with generators 2, 3. This proves that (4.2) consists of the least representable auxiliary integers in each residue modulo B .

For $n \geq 3$, one has $A > 5(k-1)$. For $k = 1$ the assertion is immediate. For $k = 2$ the smallest relevant n is 3, giving $A = 7 > 5$; for $k = 3$ it is 8, giving $A = 68 > 10$; and for $k = 6$ it is 5, giving $A = 55 > 25$. In each of these rows A increases with n . Apply Lemma 4.1 with $H = 5$ and $t = 2$, using that B is odd. This proves (4.4) for $n \geq 3$.

At $n = 2$, the parameters are $(A, B, C) = (5, 7, 15)$ and $\mathcal{L}_2 = \{0, 2, 3, 4, 5, 6, 8\}$. The corresponding weights are 0, 15, 5, 30, 20, 10, 25, exactly the set $\{0, 5, 10, 15, 20, 25, 30\} = \text{Ap}(\langle 5, 7 \rangle; 7)$. Since $15 = 3 \cdot 5$, this is also the required Apéry set for $\langle 5, 7, 15 \rangle$. This explicit check covers the only value for which the sufficient inequality in Lemma 4.1 fails. Finally apply Lemma 3.1. $\square$

5 Taking the maximum and explaining all six exceptions

Write $M = \max_{L \in \mathcal{L}_n} w(L)$. The Frobenius number of the reduced semigroup is $M - B$, and (3.6) gives

$$F_n = de(M - B) + (e - 1)a + (d - 1)c. \quad (5.1)$$

The only task remaining for the largest gap is to identify M .

Table 4: Exact maximum candidates; both candidates are needed at the small exceptional indices.

$n \bmod 6$	M
0, 4	$A(B - 1)$
1	$\max\{A(B + 1)/2, A(B - 5)/2 + C\}$
2	$\max\{A(B - 1)/3 + C, A(B - 7)/3 + 2C\}$
3	$\max\{A(B - 1)/2, A(B - 3)/2 + C\}$
5	$\max\{A(B - 1)/6, A(B - 7)/6 + 5C\}$

To verify Table 4, fix v . The weight $Au + Cv$ increases with u , so only the largest allowed u matters. Since $B \equiv 1 \pmod{k}$, that largest value, before the possible extra auxiliary integer $B + 1$, is

$$U_v = \frac{B - 1}{k} - \left\lceil \frac{\ell v}{k} \right\rceil. \quad (5.2)$$

For residue 1, the extra $B + 1$ has $v = 0$ and supplies the first candidate. For residue 2, it has $v = 1$ and supplies the first candidate; the $v = 0$ candidate is smaller by C . For residue 5, all $v = 1, \dots, 5$ have $u = (B - 7)/6$, so only $v = 5$ competes with $v = 0$. The other rows are immediate. This derives every candidate rather than inferring it from data.

The second candidate minus the first is, respectively,

$n \bmod 6$	difference	first candidate wins from
1	$C - 3A = (5B - 3A)/2$	$n = 7$
2	$C - 2A = (5B - 4A)/3$	$n = 8$
3	$C - A = (5B - A)/2$	$n = 9$
5	$5C - A = (25B - A)/6$	$n = 29$.

(5.3)

Here “from” means along the indicated residue class. The signs reduce to quadratic inequalities. In the first row the numerator $5B - 3A$ is $-2n^2 + 9n + 15$; in the second, $5B - 4A = -4n^2 + 8n + 15$; in the

third, $5B - A = (-2n^2 + 9n + 15)/3$; and in the fourth, $25B - A = -2n^2 + 49n + 75$. These are negative at and beyond the listed thresholds. They are positive at the earlier allowed indices, namely 2, 3, and 5, 11, 17, 23 in the last three rows. There are no earlier allowed indices in the first row because $n \geq 2$.

Substituting the first candidates into (5.1) and simplifying gives the four polynomials in Table 1. At an exceptional index the correction is de times the positive difference in (5.3). This gives exactly Table 2, proving the formula and the sharp cutoff in Theorem 2.1.

Why early interpolation is hazardous. The exceptional behavior is caused by a genuine change of maximizing Apéry element, not by an indexing error. In residue class five the exceptional branch persists through $n = 23$. A polynomial fitted only to small values in that residue can therefore look consistent while describing the wrong eventual branch. The normal form identifies the exact competition and proves where it ends.

6 All gaps, constructive membership, and a worked example

6.1 Membership without constructing the Apéry set

The normal form yields a bounded-operation membership procedure. Given an integer x , choose

$$\begin{aligned} 0 \leq j < d, \quad jc \equiv x \pmod{d}, \quad y &= \frac{x - jc}{d}, \\ 0 \leq i < e, \quad iA \equiv y \pmod{e}, \quad z &= \frac{y - iA}{e}. \end{aligned} \tag{6.1}$$

The congruences have unique solutions because of the coprimality proved in Lemma 3.1. A congruence modulo one simply selects zero. Let L_0 be the least residue of $2z$ modulo B . Set $L = L_0$, except that $L = B + 1$ when $L_0 = 1$ and $n \equiv 1, 2 \pmod{6}$.

Corollary 6.1 (Membership and representation certificate). *With the definitions above,*

$$x \in S_n \iff z \geq w(L). \tag{6.2}$$

When membership holds, $q = (z - w(L))/B$ is a nonnegative integer and

$$x = (i + eu(L))a + qb + (j + dv(L))c. \tag{6.3}$$

Proof. The decoding recovers the unique offset class in (3.4), even when x or an intermediate quotient is negative. The congruences in (3.9) imply $2w(L) \equiv L \equiv 2z \pmod{B}$, so $w(L)$ is the least reduced representative in the residue of z . Since 2 is invertible modulo B , the difference $z - w(L)$ is a multiple of B . Theorem 4.2 now proves (6.2); substituting the definitions proves (6.3). All its coefficients are nonnegative. $\square$

For fixed n , the two modular inverses can be precomputed. Each subsequent membership query uses a bounded number of exact arithmetic operations, independent of b and F_n . This is an arithmetic-operation statement, not a claim that arbitrarily large integers take constant bit time. The program `representation` in `verify.py` implements the formula and returns either a nonnegative coefficient triple or `None`.

6.2 The complete generating function of the semigroup

Writing $R_m(t) = 1 + t + \dots + t^{m-1}$, (4.5) gives

$$H_{S_n}(z) = \frac{R_e(z^a)R_d(z^c)}{1 - z^b} \sum_{L \in \mathcal{L}_n} z^{de w(L)}. \tag{6.4}$$

Consequently the *gap polynomial* is exactly

$$\sum_{t \in \text{Gap}(S_n)} z^t = \frac{1}{1 - z} - H_{S_n}(z). \tag{6.5}$$

The right side is a polynomial despite its displayed rational form: the residue description proves that all sufficiently large coefficients cancel. This is a full description of the complement, not just an estimate for its largest element.

6.3 The exceptional case $n = 5$

Here $(a, b, c) = (55, 91, 140)$, $(d, e) = (1, 7)$, $(k, \ell) = (6, 1)$, and $(A, B, C) = (55, 13, 20)$. The relation is $6 \cdot 20 - 55 = 5 \cdot 13$. For $L = 0, \dots, 12$, the reduced weights are

$$0, 20, 40, 60, 80, 100, 55, 75, 95, 115, 135, 155, 110.$$

Their maximum is 155 at $L = 11$, not the eventual-branch candidate 110 at $L = 12$. Hence

$$F(\langle 55, 13, 20 \rangle) = 155 - 13 = 142, \quad F_5 = 7 \cdot 142 + 6 \cdot 55 = 1324.$$

The largest original Apéry representative is $7 \cdot 155 + 6 \cdot 55 = 1415$, and subtracting $b = 91$ gives the same Frobenius number. The reduced sum is 1040, so its genus is $1040/13 - 6 = 74$; the original genus is $7 \cdot 74 + (330 - 7 + 1)/2 = 680$. The entire exception is therefore explained by a single visible change of maximum in a thirteen-element set.

7 The genus, symmetry, and every gap moment

7.1 A genus formula with no exceptions

Let $g_n = \# \text{Gap}(S_n)$, and define the six-periodic sequence

$$J_n = (0, 9, 8, 9, 0, 5)_{n \bmod 6}. \quad (7.1)$$

Theorem 7.1 (Exact genus). *For every $n \geq 2$,*

$$g_n = \frac{n(n+1)(4n^3 + 28n^2 + 73n + 102) - 5J_n(n+1)(n+2)}{72}. \quad (7.2)$$

There are no exceptional indices. In particular,

$$g_n = \frac{n^5}{18} + \frac{4n^4}{9} + \frac{101n^3}{72} + \frac{175 - 5J_n}{72}n^2 + \frac{102 - 15J_n}{72}n - \frac{5J_n}{36}. \quad (7.3)$$

Thus the first three coefficients are independent of the residue class.

Proof. Put $\eta = 1$ in residues 1, 2 and $\eta = 0$ otherwise. Directly from (4.2),

$$\sum_{L \in \mathcal{L}_n} L = \frac{B(B-1)}{2} + B\eta.$$

The sum $V = \sum_{L \in \mathcal{L}_n} v(L)$ is

$$\begin{array}{c|cccccc} n \bmod 6 & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline V & 0 & (B-3)/2 & B-2 & (B-1)/2 & 0 & 5(B-1)/2. \end{array} \quad (7.4)$$

For example, in residue 2 the unmodified interval contributes $B-1$; removing $L=1$ removes $v=2$, and adding $L=B+1$ adds $v=1$. In residue 1 the analogous change removes one. The other rows just sum complete periods of $v(L)$.

Using (4.3) and (3.2), the reduced genus is

$$g(T) = \frac{A}{k} \left(\frac{B-1}{2} + \eta \right) + \frac{5}{k}V - \frac{B-1}{2}. \quad (7.5)$$

Insert this into (3.7) and use (3.8). The result in each of the six rows is (7.2). All identities remain valid at $n=2$ by the explicit Apéry set already proved in Theorem 4.2. $\square$

Corollary 7.2 (Exact symmetry classification). *The semigroup S_n is symmetric if and only if*

$$n = 2 \quad \text{or} \quad n \equiv 0, 4 \pmod{6}. \quad (7.6)$$

For every other $n \geq 2$, it is nonsymmetric, although $g_n/F_n \rightarrow 1/2$.

Proof. A numerical semigroup is symmetric when for every integer t exactly one of $t, F - t$ belongs to it. Pairing the integers in $[0, F]$ shows the standard equivalence with $2g = F + 1$: both members of a pair cannot be in the semigroup, and equality means there are no pairs with both members gaps. Here direct subtraction gives

$$2g_n - F_n - 1 = \begin{cases} 0, & n \equiv 0, 4 \pmod{6}, \\ \frac{5n(n+1)(n+2)}{6} - \varepsilon_n, & n \equiv 1, 3 \pmod{6}, \\ \frac{5(n+1)(n+2)(2n-1)}{6} - \varepsilon_n, & n \equiv 2 \pmod{6}, \\ \frac{25(n+1)^2(n+2)}{18} - \varepsilon_n, & n \equiv 5 \pmod{6}. \end{cases} \quad (7.7)$$

At the exceptional indices 2, 3, 5, 11, 17, 23, the actual defects are 0, 10, 35, 988, 5415, 17600. Outside them the nonzero branches are positive. This proves the classification. The asymptotic ratio follows from the leading terms of (2.6) and (7.3). $\square$

7.2 A closed formula for every power sum

For an integer $p \geq 0$, define

$$M_p(n) = \sum_{t \in \text{Gap}(S_n)} t^p, \quad M_0(n) = g_n.$$

Let $B_j(X)$ be the Bernoulli polynomials in the convention

$$\frac{ue^{Xu}}{e^u - 1} = \sum_{j \geq 0} B_j(X) \frac{u^j}{j!}, \quad \beta_j = B_j(0), \quad \beta_1 = -\frac{1}{2}.$$

Proposition 7.3 (Apéry–Bernoulli moment formula). *For a numerical semigroup S and $m \in S \setminus \{0\}$,*

$$\sum_{t \in \text{Gap}(S)} t^p = \frac{m^p}{p+1} \sum_{w \in \text{Ap}(S; m)} B_{p+1}(w/m) - \frac{\beta_{p+1}}{p+1}. \quad (7.8)$$

In particular, for $S = S_n$, the representatives in Theorem 4.2 give a finite exact formula for every $M_p(n)$.

Proof. Write $w_r = r + mq_r$. Summing powers in the arithmetic progression (3.1) gives

$$\sum_{j=0}^{q_r-1} (r + mj)^p = \frac{m^p}{p+1} (B_{p+1}(w_r/m) - B_{p+1}(r/m)).$$

The Bernoulli multiplication identity $\sum_{r=0}^{m-1} B_{p+1}(r/m) = m^{-p} \beta_{p+1}$ follows immediately by summing the defining generating functions. Now sum over r . $\square$

The classical power-sum identity in this proposition is not a novelty claim. What makes it effective here is the complete uniform description of its Apéry input. The next theorem gives a global consequence for all moment orders, without requiring each polynomial to be separately guessed.

Theorem 7.4 (Exact all-orders moment quasipolynomials). *For every fixed integer $p \geq 0$, $M_p(n)$ is a quasipolynomial of period dividing six for all $n \geq 2$. Each residue-class polynomial has degree $5(p+1)$ and leading coefficient*

$$\frac{1}{9^{p+1}(p+1)(p+2)}. \quad (7.9)$$

Thus every moment has a complete finite asymptotic expansion with periodic coefficients. No exceptional corrections are needed for the moments.

Proof. First prove quasipolynomiality. Fix $n \bmod 6$, so k, ℓ are constant. For each $v = 0, \dots, k-1$, the representable auxiliary integers in $[0, B-1]$ have $L = ku + \ell v$, with $0 \leq u \leq U_v$ from (5.2). In residues 1, 2, add the single extra weight at $B+1$. Consequently, for any $j \geq 0$,

$$\sum_{L \in \mathcal{L}_n} w(L)^j = \sum_{v=0}^{k-1} \sum_{u=0}^{U_v} (Au + Cv)^j + \eta w(B+1)^j. \quad (7.10)$$

When $\eta = 0$ the last term is omitted. Every U_v , d , and e is an affine function of n on the fixed residue class. All their relevant values and all displayed ranges are valid for the allowed $n \geq 2$. Expanding powers and summing ordinary powers proves that every power sum of the full representatives $de w + ia + jc$ is a polynomial in n on that residue class. Formula (7.8) therefore expresses $M_p(n)$ as a *rational function* of n with rational coefficients.

There is a useful elementary integrality argument to remove the possible denominator. If $R(q) \in \mathbb{Q}(q)$ is integral for all sufficiently large integers q , perform polynomial division $R = Q + U$, with $Q \in \mathbb{Q}[q]$ and $U(q) \rightarrow 0$. A fixed integer $D > 0$ clears all coefficients of Q , so $U(q) \in D^{-1}\mathbb{Z}$ at those integer arguments. For large q , $|U(q)| < 1/D$, hence $U(q) = 0$. A rational function with infinitely many zeros is identically zero. Thus R is a polynomial. Apply this to $n = 6q + r$ and $M_p(6q + r)$. Since the rational formula was already valid for every allowed $n \geq 2$, its polynomial simplification is valid at all those indices, not merely eventually.

The degree and leading coefficient follow from the independently proved limit law in Section 8, which uses only the Apéry normal form and the Frobenius formula, not the present quasipolynomiality claim. Specifically it gives

$$M_p(n) \sim \frac{F_n^{p+1}}{(p+1)(p+2)} \sim \frac{n^{5p+5}}{9^{p+1}(p+1)(p+2)}.$$

A polynomial with this positive asymptotic leading term has exactly the stated degree and leading coefficient in every residue class. $\square$

For example, the first gap-power sum starts

$$M_1(n) = \frac{n^{10}}{486} + \frac{29n^9}{972} + \frac{395n^8}{1944} + c_{n \bmod 6} n^7 + O(n^6), \quad (7.11)$$

where

$$(c_0, c_1, c_2, c_3, c_4, c_5) = \frac{1}{3888} (3455, 3185, 3215, 3185, 3455, 3305).$$

Appendix A gives all terms exactly. The program `derive_moments.py` implements (7.10), the rectangular lift, and (7.8); it generates the polynomial for a requested order by symbolic summation, not interpolation. Exact order-one and order-two outputs accompany the report.

8 A general gap-limit principle and the triangular law

8.1 Transferring a limit of Apéry sets

Let S_j be a sequence of numerical semigroups with Frobenius numbers $F_j \rightarrow \infty$. Choose any $m_j \in S_j \setminus \{0\}$ such that $m_j/F_j \rightarrow 0$. Define the normalized empirical Apéry measure

$$\nu_j = \frac{1}{m_j} \sum_{w \in \text{Ap}(S_j; m_j)} \delta_{w/F_j}.$$

Its support is contained in $[0, 1 + m_j/F_j]$, since the maximum Apéry element is $F_j + m_j$.

Theorem 8.1 (Gap-limit transfer principle). *Suppose ν_j converges weakly to a probability measure ν on $[0, 1]$. Put $\mu = \int t \, d\nu(t)$. Then $\mu \geq 1/2$, and the cumulative distribution of a uniformly chosen gap divided by F_j converges uniformly on $[0, 1]$ to*

$$G(x) = \frac{\int_0^1 \min(x, t) \, d\nu(t)}{\mu}. \quad (8.1)$$

The limit is absolutely continuous and has a nonincreasing density

$$g(x) = \frac{\nu((x, 1])}{\mu} \quad \text{for almost every } x \in (0, 1), \quad 0 \leq g(x) \leq 2. \quad (8.2)$$

Thus its cumulative distribution is concave.

Proof. For $0 \leq x \leq 1$, count the gaps in each residue class using (3.1). The number in residue r not exceeding xF_j differs from $\min(xF_j, w_r)/m_j$ by at most an absolute constant. Hence, uniformly in x ,

$$\#\{t \in \text{Gap}(S_j) : t \leq xF_j\} = F_j \int \min(x, t) \, d\nu_j(t) + O(m_j). \quad (8.3)$$

At $x = 1$ this, or the exact genus formula, gives $g(S_j)/F_j \rightarrow \mu$. Pairing t and $F_j - t$ shows $g(S_j) \geq (F_j + 1)/2$, so $\mu \geq 1/2$.

The integrands $\min(x, t)$ are uniformly bounded and uniformly Lipschitz in both variables on the relevant compact interval. Weak convergence, a finite mesh in x , and the Lipschitz bound therefore give uniform convergence of their integrals. Divide (8.3) by the genus to obtain (8.1). Finally,

$$\int \min(x, t) \, d\nu(t) = \int_0^x \nu((u, 1]) \, du.$$

This identity follows by integrating the indicator $\mathbf{1}_{\{u < t\}}$ and interchanging two nonnegative integrals. It proves absolute continuity, the density formula, monotonicity, and the bound $g \leq 1/\mu \leq 2$. $\square$

The transfer principle gives a useful necessary shape constraint for limiting gap distributions whenever a semigroup modulus is negligible relative to the Frobenius number. By compactness, every subsequential limit can be treated after extracting a subsequence of the Apéry measures. This concerns the *sizes of the gaps*; it is not a statement about distributions of factorization lengths or about random choice of a numerical semigroup.

8.2 Uniform Apéry distribution for square-pyramidal generators

Theorem 8.2 (Quantitative triangular gap law). *Let X_n be uniformly distributed on $\text{Gap}(S_n)$. There is an absolute constant C such that for every $n \geq 2$,*

$$\sup_{0 \leq x \leq 1} \left| \mathbb{P}\left(\frac{X_n}{F_n} \leq x\right) - (2x - x^2) \right| \leq \frac{C}{n}. \quad (8.4)$$

In particular, X_n/F_n converges to the probability density $2(1 - x)$ on $[0, 1]$. For each fixed integer $p \geq 0$,

$$\mathbb{E} \left[\left(\frac{X_n}{F_n} \right)^p \right] = \frac{2}{(p+1)(p+2)} + O_p(n^{-1}), \quad (8.5)$$

$$M_p(n) = \frac{n^{5p+5}}{9^{p+1}(p+1)(p+2)} + O_p(n^{5p+4}). \quad (8.6)$$

The limiting normalized mean and variance are $1/3$ and $1/18$.

Proof. Use the original modulus $m = b$, so $m = O(n^3)$ and $m/F_n = O(n^{-2})$. Write a full representative as $W = de w(L) + ia + jc$. Define the residue-independent polynomial

$$T_n := \frac{deAB}{k} = \frac{n(n+1)(n+2)(2n+1)(2n+3)}{36}. \quad (8.7)$$

Then (4.3) gives

$$W = T_n \frac{L}{B} + \frac{5deB}{k} v(L) + ia + jc.$$

Here $v(L) \leq 5$, $d, e = O(n)$, $A, C = O(n^2)$, $B \asymp n$, and $a, c = O(n^3)$, uniformly over all six residues. Thus the last three terms are $O(n^4)$, while $F_n \asymp n^5$ and $T_n/F_n = 1 + O(n^{-1})$. Since $0 \leq L \leq B + 1$, it follows that

$$\left| \frac{W}{F_n} - \frac{L}{B} \right| = O(n^{-1}) \quad (8.8)$$

uniformly over *every* representative in (4.5).

Each L occurs with exactly the same multiplicity de . The empirical measure on the numbers L/B differs from the uniform grid $0, 1/B, \dots, (B-1)/B$ by at most one replacement. Its integral against a bounded Lipschitz function therefore differs from the integral against uniform measure on $[0, 1]$ by $O(1/B) = O(1/n)$. For the uniformly bounded 1-Lipschitz functions $t \mapsto \min(x, t)$, (8.8) gives, uniformly in $x \in [0, 1]$,

$$\frac{1}{b} \sum_{W \in \text{Ap}(S_n; b)} \min(x, W/F_n) = \int_0^1 \min(x, t) dt + O(n^{-1}) = x - \frac{x^2}{2} + O(n^{-1}). \quad (8.9)$$

Apply (8.3). The genus satisfies $g_n/F_n = 1/2 + O(n^{-1})$ (indeed the exact formulas give a stronger error). Division gives (8.4) for all sufficiently large n . Enlarging the constant covers the finitely many smaller values.

Integrating against x^p yields $\int_0^1 2x^p(1-x) dx = 2/((p+1)(p+2))$. The uniform cumulative-distribution error gives the same $O_p(n^{-1})$ error for each fixed moment; at $p = 0$ the identity is exact. Multiply by $g_n F_n^p$ and use the leading Frobenius and genus terms to prove (8.6). The mean and variance follow from the first two normalized moments. $\square$

The limit theorem is not based on a histogram. It follows from a uniform comparison of every normalized Apéry representative with a regular grid. The $O(n^{-1})$ estimate is deliberately robust; determining optimal second-order corrections is a separate research question.

9 The rational generating function and the minimal recurrence

Let $\Theta = x d/dx$ and set

$$U(x) = \frac{1}{1-x}, \quad H(x) = \frac{3x}{1-x^2} + \frac{2x^2}{1-x^3}, \quad (9.1)$$

so that $H(x) = \sum_{n \geq 0} h_n x^n$. The correction polynomial is

$$E(x) = 10x^2 + 40x^3 + 315x^5 + 1612x^{11} + 3135x^{17} + 2400x^{23}. \quad (9.2)$$

Theorem 9.1 (Exact generating function and minimal denominator). *The ordinary generating function $\Phi(x) = \sum_{n \geq 2} F_n x^n$ is*

$$\begin{aligned} \Phi(x) = & \frac{1}{36} (4\Theta^5 + 32\Theta^4 + 101\Theta^3 + 175\Theta^2 + 102\Theta - 36)U(x) \\ & - \frac{5}{36} (\Theta + 1)(\Theta + 2)(2\Theta + 3)H(x) + 1 + 2x + E(x). \end{aligned} \quad (9.3)$$

In reduced form its denominator is exactly

$$D(x) = (1-x)^6(1+x)^4(1+x+x^2)^4. \quad (9.4)$$

Its numerator has degree 41. The least eventual quasiperiod of F_n is six, and its minimal eventual homogeneous constant-coefficient recurrence has order 18.

Proof. Multiplication of the n th coefficient by n is the action of Θ . Thus (9.3) follows termwise from (2.4) and (2.5). The extended quasipolynomial has coefficients $-1, -2$ at $n = 0, 1$, respectively, explaining the terms $1 + 2x$.

The U term has a pole of order six at $x = 1$, with nonzero leading coefficient. The H term has order at most four there and cannot cancel it. At $x = -1$, the first summand of H has a nonzero simple pole; applying the degree-three differential operator in (9.3) raises it to a pole of order four, with nonzero leading term. The second summand of H is regular there. At each primitive cube root of unity the second summand similarly produces a pole of order four, while the first is regular. There are no other poles, and the added polynomials do not affect them. This proves that (9.4) is the exact reduced denominator.

An eventual quasipolynomial of period q has poles only at q th roots of unity. The poles at -1 and the primitive cube roots force $6 \mid q$; period six is already supplied by (2.5). The pole multiplicities likewise force every eventual recurrence denominator to contain (9.4); its degree is $6 + 4 + 8 = 18$. Conversely this denominator provides such a recurrence.

The highest-degree polynomial correction is $2400x^{23}$. Multiplying it by $D(x)$, whose degree is 18 and leading coefficient is one, produces the nonzero coefficient $2400x^{41}$. All other terms have lower degree after multiplication by D , proving the numerator degree. $\square$

Writing $D(x) = \sum_{j=0}^{18} d_j x^j$, the coefficients in increasing order of j are

$$(d_0, \dots, d_{18}) = (1, 2, -1, -8, -9, 6, 23, 16, -14, -32, -14, 16, 23, 6, -9, -8, -1, 2, 1). \quad (9.5)$$

Therefore the explicit recurrence is

$$\boxed{\sum_{j=0}^{18} d_j F_{n-j} = 0 \quad (n \geq 42)}. \quad (9.6)$$

At $n = 41$ the left side is 2400, so the starting point is sharp for this recurrence. The absence of primitive sixth-root poles is worth noting: period six arises by combining frequencies of periods two and three, not by requiring all sixth roots. A generic denominator $(1 - x^6)^6$ would therefore be far from minimal.

10 All-orders algebraic inversion

10.1 The continuous branches

For $h \in \{0, 2, 3, 5\}$, let $N_h(y)$ be the large positive solution of $P_h(N_h(y)) = y$. Set

$$t = (9y)^{1/5}, \quad R_h(u) = 1 + 8u + \frac{101 - 10h}{4}u^2 + \frac{175 - 45h}{4}u^3 + \frac{102 - 65h}{4}u^4 - \frac{36 + 30h}{4}u^5. \quad (10.1)$$

Then $9P_h(N) = N^5 R_h(1/N)$.

Theorem 10.1 (Convergent Puiseux inverse). *For sufficiently large positive y ,*

$$N_h(y) = t - \frac{8}{5} + \sum_{m \geq 1} c_m(h) t^{-m}, \quad c_m(h) = -\frac{1}{m} [u^{m+1}] R_h(u)^{m/5}. \quad (10.2)$$

This series is convergent, not merely a formal asymptotic expansion. In particular,

$$\begin{aligned} N_h(y) = t - \frac{8}{5} + \frac{50h + 7}{100t} - \frac{3(25h + 149)}{500t^2} \\ + \frac{2500h^2 - 400h + 37817}{10000t^3} - \frac{3(6250h^2 + 37125h + 17199)}{250000t^4} + O(t^{-6}). \end{aligned} \quad (10.3)$$

There is no t^{-5} term. More generally,

$$\boxed{c_{5j}(h) = 0 \quad (j \geq 1)}. \quad (10.4)$$

Proof. Put $z = 1/t$ and $u = 1/N$. The defining equation becomes

$$u = z R_h(u)^{1/5}, \quad (10.5)$$

using the analytic fifth root equal to one at $u = 0$. The implicit-function theorem gives a unique analytic solution $u(z) = z + O(z^2)$ near zero. Thus $1/u(z)$ has a convergent Laurent expansion with leading term z^{-1} .

For clarity, the coefficient calculation can be seen directly from Lagrange inversion. Write $\varphi(u) = R_h(u)^{1/5}$. For $m \geq 1$, the change of variable $z = u/\varphi(u)$ in coefficient extraction gives

$$\begin{aligned} [z^m] \frac{1}{u(z)} &= [u^{m+1}] \varphi(u)^m - [u^m] \varphi(u)^{m-1} \varphi'(u) \\ &= -\frac{1}{m} [u^{m+1}] \varphi(u)^m. \end{aligned}$$

The constant coefficient is $-\varphi'(0) = -8/5$. This proves (10.2); ordinary binomial expansion yields (10.3). For $m = 5j$, the expression $R_h(u)^j$ is a polynomial of degree $5j$, so its coefficient at u^{5j+1} vanishes. This proves (10.4) and explains the $O(t^{-6})$ remainder after the fourth inverse-power term. $\square$

The missing coefficients are a general consequence of inverting a polynomial at infinity: the same argument for a monic degree- d polynomial shows that the coefficients of t^{-dj} vanish. This is a useful Lagrange-inversion corollary, not a claim of a new general inversion theorem. For the present problem it provides an all-orders structural check on symbolic calculations.

[write] Added 2 October 2026, batch 75: an instance of repository results. The coefficient formula (10.2) is the Lagrange–Bürmann theorem `p0:thm:LB` of the transseries-and-inversion volume [7], applied to the fixed-point form (10.5) with $\varphi = R_h^{1/5}$; the proof above extracts the coefficients of the Laurent series $1/u$ by the same residue computation that proves that theorem for power series G . The lacunarity (10.4) is the degree- d corollary stated in the paragraph above. Theorem 10.1 is an instance of that apparatus, and no novelty is claimed for the method; the convergence region of Proposition 10.2 is specific to this report.

Proposition 10.2 (A uniform sufficient convergence region). *The Laurent series in (10.2) converges for real $t > 100$, uniformly over the four allowed h values. If it is truncated after $c_m(h)t^{-m}$, with $m \geq 0$ and $m = 0$ meaning $t - 8/5$, the error is at most*

$$\frac{2 \cdot 100^{m+2}}{t^{m+1}(1 - 100/t)}. \quad (10.6)$$

This is a conservative bound, not an estimate of the actual convergence radius or the actual numerical error.

Proof. On $|u| \leq 1/50$, the coefficient bounds in (10.1) give $|R_h(u) - 1| < 0.18$ and $|R'_h(u)| < 9.2$ for every allowed h . The principal fifth root is analytic on this disk. For $|z| \leq 1/100$, the map $u \mapsto z R_h(u)^{1/5}$ maps the disk strictly into itself and has derivative in modulus less than 0.03. Its iterates from zero converge uniformly, so the fixed point is analytic in z . The analytic function $V(z) = z/u(z) = R_h(u(z))^{-1/5}$ satisfies $|V(z)| < 2$. Cauchy's coefficient estimate on the circle $|z| = 1/100$, followed by a geometric-series bound, bounds the remainder of $z^{-1}V(z)$ after its z^m term by (10.6). $\square$

No logarithmic or exponentially small sectors are needed in this problem. The forward expansion is a finite exact quasipolynomial, and each inverse branch is an algebraic function with a convergent Puiseux expansion. This is stronger information than a purely formal transseries ansatz.

10.2 The exact discrete inverse

Define

$$I(y) = \max\{n \geq 2 : F_n \leq y\},$$

whenever the set is nonempty. First note that $P_h(X)$ is strictly increasing for $X \geq 2$. Indeed, its derivative is minimized among the allowed h by $h = 5$, and

$$36P'_5(X) = 20X^4 + 128X^3 + 153X^2 - 100X - 223 > 0 \quad (X \geq 2).$$

For the last inequality, $153X^2 - 100X - 223 = (X - 2)(153X + 206) + 189 > 0$. The entire sequence F_n is also strictly increasing. The finite range $2 \leq n < 24$ is checked directly from (2.5); for $n \geq 24$, Appendix B gives positive-coefficient polynomial certificates for all six adjacent differences.

Corollary 10.3 (Residue-aware inverse rounding). *For $y \geq F_{29} = 2940909$,*

$$I(y) = \max_{0 \leq r < 6} \left\{ r + 6 \left\lfloor \frac{N_{h_r}(y) - r}{6} \right\rfloor \right\}, \quad (h_0, \dots, h_5) = (0, 3, 2, 3, 0, 5). \quad (10.7)$$

Proof. For this range of y , every residue class has an eligible index at least $24 + r$. On those tails the exceptional corrections have disappeared, and the relevant continuous polynomial is strictly increasing. Its largest eligible integer in residue r is therefore exactly the lattice floor in (10.7). Taking the largest of the six candidates proves the result; smaller exceptional indices cannot affect the maximum. $\square$

A single continuous inverse followed by ordinary rounding would in general ignore the periodic correction. Formula (10.7) keeps that information. For numerical use, evaluate an inverse expansion to suggest a candidate and then certify it using exact comparisons with (2.5); near a lattice boundary, the exact comparison is essential. The included `inverse_index` routine instead uses monotone binary search and exact integer arithmetic, so it has no floating-point rounding ambiguity.

[write] **Added 2 October 2026, batch 75: an instance of repository results.** Corollary 10.3 is the residue-class case, part (4), of the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [7] with modulus $r = 6$. It is written for the largest index with $F_n \leq y$, with floors and a maximum, where that theorem treats the least index with $F_n \geq y$, with ceilings and a minimum. The warning in the paragraph above, that an approximate inverse determines the integer only away from a lattice boundary, is part (2) of that theorem, its separation condition. The generic arithmetic of parts (1), (2) and (4) is formalized in Lean as `Fabius.staircase_ceil`, `Fabius.staircase_separation` and `Fabius.isLeast_residue_class` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those are statements about an arbitrary monotone interpolation, not about F_n , and give no formal status to this corollary. No novelty is claimed for the rounding rule.

11 Reproducibility and limits of the computational audit

The accompanying code uses two genuinely different descriptions. The new normal form constructs a short reduced Apéry set and lifts it. An independent Dijkstra implementation finds least representatives as shortest path distances in the residue graph: a generator gives an edge of that weight from r to $r + g$ modulo the chosen modulus. It knows nothing about the quasipolynomial or the maximum-candidate table.

The random membership samples use a fixed seed, so they are reproducible. The boundary checks are deterministic. The main verification run and the symbolic derivations for moment orders one and two all passed. The exact console outputs are preserved in the package. The programs generate a CSV with F_n , genus, and symmetry defect through $n = 1000$.

To reproduce the main audit in a Python environment with SymPy installed, run

```
python verify.py -symbolic
python derive_moments.py -power 1
python derive_moments.py -power 2
```

The first command without `-symbolic` uses only the Python standard library. The manuscript compiles with two runs of `pdflatex`.

Table 5: Executed exact checks. These supplement the infinite proofs.

Range	Check
$n = 2, \dots, 35$	All 34 values displayed in the inspected OEIS entry.
$n = 2, \dots, 1000$	Reduced Apéry set against independent Dijkstra; gcd identities; Frobenius and genus lifting.
$n = 2, \dots, 60$	Frobenius and genus from the original three generators, with no gcd reduction.
$n = 2, \dots, 30$	Every lifted Apéry representative against Dijkstra modulo the original b .
$n = 2, \dots, 200$	Constructive representation checks, largest-gap and conductor boundary checks; direct membership comparisons through $n = 30$.
$n = 2, \dots, 12$	Gap moments of orders 0, 1, 2, 3 against direct dynamic programming enumeration of the gaps.
$n = 42, \dots, 1000$	The order-18 recurrence; the boundary residual at $n = 41$ is checked to be 2400.
$n = 2, \dots, 300$	Exact inverse at the left and right ends of each interval $[F_n, F_{n+1})$.
Symbolic	Six residue identities for the reductions and genus; positivity certificates; denominator and coprimality; the first inverse coefficients.

[write] **Added 2 October 2026, batch 75.** In the repository the programs live in `code/` and their recorded outputs in `data/`. Run as printed, `verify.py` writes `data.csv` into its own directory, and `derive_moments.py` imports `verify`; the report's README gives a recipe that runs both on a scratch copy and leaves the shipped files untouched. On 2 October 2026 that recipe reproduced `data.csv` byte for byte, `verification.txt` apart from its seconds line, and both moment files and their outputs apart from line endings.

What was not established by computation. No finite range, however large, proves the eventual formula, the completeness of the exception list, or the limiting distribution. Those claims rest on the proofs above. No machine-checked theorem prover was run for this report. No OEIS edit or GitHub commit was submitted. The mathematical claims are therefore conventional proved statements in an unrefereed manuscript, with reproducible computational support and an explicitly limited novelty assessment.

12 Further research questions

The following directions concern extensions not proved in this report. They are presented as research questions, not as conjectures whose truth has been silently assumed.

1. Higher sums of powers. Replace s_n by $s_n^{(r)} = \sum_{j=1}^n j^r$ for a fixed $r \geq 3$. Three consecutive values still have gcd one, because consecutive differences are coprime powers. Determine the adjacent-gcd structure, minimal quasiperiods, degrees, and effective cutoffs of their Frobenius quasipolynomials. The present argument suggests searching first for a reduced relation with bounded coefficients, but it does not predict a universal degree without further work.

2. More consecutive generators. Determine exact Frobenius formulas for $\langle s_n, s_{n+1}, \dots, s_{n+r} \rangle$ with fixed $r \geq 3$. Does adding a fourth generator lower the growth exponent, alter only the leading constant, or create new residue-dependent regimes? A useful solution should include an explicit normal form rather than only invoke general eventual quasipolynomiality.

3. A classification of bounded-relation families. Characterize polynomial triples which, after gcd reductions, satisfy a relation $kC - \ell A = HB$ with bounded k, ℓ, H , together with compatible residue scalings. Develop an effective criterion that recognizes when Lemma 4.1 yields the full Apéry set. This would turn the successful structure here into a systematic proof-generation method.

4. Optimal and second-order gap limits. Determine the optimal uniform rate in (8.4). More finely, does a residue-dependent second-order expansion exist for the cumulative gap count, away from boundary layers and arithmetic discontinuities? The explicit rectangular lift exposes several scales, of orders n^{-1} and n^{-2} after normalization, whose interactions should be analyzed rather than averaged away.

5. Realizable limiting gap densities. Theorem 8.1 forces a nonincreasing density bounded by two when a semigroup modulus is $o(F)$. Which such densities actually arise from numerical semigroups? What additional restrictions does addition closure place on the candidate Apéry limit measure? In particular, characterize the circumstances giving uniform, triangular, or piecewise-polynomial gap limits.

6. Minimal recurrences for higher moments. For each fixed p , determine the exact pole orders of the generating function of $M_p(n)$, not just the generic bound supplied by its degree and period. The first three coefficients of g_n and of $M_1(n)$ are residue independent, so systematic cancellations occur. A theorem giving the minimal recurrence order as a function of p would sharpen the all-orders moment result.

7. The rest of the semigroup invariants. Use the explicit Apéry order to determine pseudo-Frobenius numbers, semigroup type, and minimal defining relations for all six residue classes. The symmetry classification is only the first structural invariant. It remains to identify how the small maximum-switching exceptions interact with these finer algebraic invariants.

8. Fast rank and selection among the gaps. Membership has a bounded-operation description, but a direct use of (3.1) for gap rank still scans many residues. Derive a compact floor-sum algorithm for counting gaps below an arbitrary threshold and for selecting the j th gap. A useful target is logarithmic-depth integer arithmetic after parameter preprocessing, with explicit certificates for the returned rank or gap.

9. Certified inverse evaluation. Find the actual nearest complex singularity of each algebraic inverse branch, replacing the conservative convergence region in Proposition 10.2. Combine this with interval arithmetic to obtain short, guaranteed inverse evaluations even when y is close to a residue-lattice boundary. The exact polynomial comparisons provide an independent correctness check.

10. Formalization and an OEIS proof artifact. Formalize the rectangular lifting lemma and the least auxiliary residue principle once, and specialize them to the six rows by exact arithmetic. A desirable deliverable is a checked theorem giving both membership certificates and the largest-gap formula, not only a finite table. Only after independent review should the exact formula and its references be proposed as an OEIS update. The separate `oeis_submission.txt` file is a draft for that purpose, not a record of a submitted change.

A The exact sum of the gaps

For every $n \geq 2$, the first gap moment is given by the following four polynomials. As with the genus, no exceptional correction is necessary. For $n \equiv 0, 4 \pmod{6}$,

$$M_1(n) = \frac{n(n+1)(n+2)(2n+1)}{3888} (4n^6 + 44n^5 + 227n^4 + 775n^3 + 1848n^2 + 2853n + 2322). \quad (\text{A.1})$$

For $n \equiv 1, 3 \pmod{6}$,

$$M_1(n) = \frac{(n+1)(n+2)(n+3)}{3888} (8n^7 + 68n^6 + 294n^5 + 625n^4 + 841n^3 + 441n^2 - 576n + 405). \quad (\text{A.2})$$

For $n \equiv 2 \pmod{6}$,

$$M_1(n) = \frac{(n+1)(n+2)(2n+5)}{3888} (4n^7 + 36n^6 + 159n^5 + 371n^4 + 548n^3 + 337n^2 - 474n + 96). \quad (\text{A.3})$$

For $n \equiv 5 \pmod{6}$,

$$M_1(n) = \frac{(n+1)(n+2)}{3888} (8n^8 + 92n^7 + 498n^6 + 1627n^5 + 3446n^4 + 5079n^3 + 4697n^2 + 1872n + 1425). \quad (\text{A.4})$$

Here is a compact derivation sufficient to reproduce every coefficient. For $m = b$ and $\mathcal{A} = \text{Ap}(S_n; b)$, the $p = 1$ case of (7.8) reads

$$M_1(n) = \frac{1}{2b} \sum_{W \in \mathcal{A}} W^2 - \frac{1}{2} \sum_{W \in \mathcal{A}} W + \frac{b^2 - 1}{12}. \quad (\text{A.5})$$

Expand $W = de w + ia + jc$ and sum over the rectangular ranges. If $T_j = \sum_{L \in \mathcal{L}_n} w(L)^j$, then for any $j \geq 0$,

$$\sum_{W \in \mathcal{A}} W^j = \sum_{j_1 + j_2 + j_3 = j} \binom{j}{j_1, j_2, j_3} (de)^{j_1} T_{j_1} a^{j_2} c^{j_3} \left(\sum_{i=0}^{e-1} i^{j_2} \right) \left(\sum_{v=0}^{d-1} v^{j_3} \right). \quad (\text{A.6})$$

Use (7.10) for T_0, T_1, T_2 , with $\sum_{u=0}^q u = q(q+1)/2$ and $\sum_{u=0}^q u^2 = q(q+1)(2q+1)/6$. Substitution into (A.5) yields (A.1)–(A.4). The accompanying symbolic script performs exactly these operations and checks the results against independently enumerated gap sums. The same multinomial identity is used for every higher moment.

B Positive-coefficient monotonicity certificates

For $r = 0, \dots, 5$, put $n = 24 + r + 6q$ with $q \geq 0$. The exact difference $F_{n+1} - F_n$ is

$$720q^4 + A_r q^3 + B_r q^2 + C_r q + D_r,$$

where Table 6 gives positive integer coefficients. Thus all these differences are positive for $q \geq 0$. The remaining consecutive differences, from $n = 2$ through $n = 23$, are positive by direct evaluation of (2.5).

Table 6: Exact monotonicity certificates on the six tails.

r	A_r	B_r	C_r	D_r
0	12348	79366	226585	242435
1	13068	88870	268396	303739
2	13428	93844	291280	338798
3	14148	104188	340807	417830
4	14148	104152	340422	416808
5	15228	120646	424390	559300

C Provenance and package contents

The package contains the manuscript source and compiled PDF, the main exact verification program and its executed output, a symbolic moment generator, exact first- and second-moment polynomial files and their outputs, a CSV of values through $n = 1000$, a machine-readable numerator and denominator for $\Phi(x)$, a draft OEIS contribution, and a README with reproduction instructions. No external paper PDFs or font files are bundled.

The source inspection was performed for this report on October 1, 2026. The relevant negative finding is limited: the inspected A069762 entry did not supply the formula, and the cited primary-source search did not locate it. This report should therefore be cited as an unrefereed explicit derivation until its proofs and literature status have been independently reviewed. Its exact assertions do not depend on any empirical extrapolation or any unproved result in ProveIt.

[write] **Added 2 October 2026, batch 75.** As placed in ProveIt the package is split into `code/` and `data/`; the compiled PDF described above was not shipped (this report’s PDF is rebuilt from the present source), and the delivery README was replaced by a report README. Every program and data file is byte-identical to the delivery.

References

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- [7] *Transseries and inversion*, a volume of the ProveIt repository: `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (theorems `p0:thm:LB` and `p0:thm:staircase`). Added in writing, batch 75.
- [8] *Leaves of large type in the numerical-semigroup tree*, a report of the ProveIt research-report collection: `SetTheory/Cardinals/docs/reports/enumerative-combinatorics/numerical-semigroup-leaf-types`. Added in writing, batch 75.